

Buy one get one free? Sequence-level information from single-cell RNA-seq

MIG Journal Club: MALVA for sequence pattern discovery from scRNA-seq

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8 Oct 2026

The question: Which cells contain this sequence?

Example: *DNMT3A* R882H (27 bases)

CGTCTCCAACATGAGCCGCTTGGCGAG (usual)

CGTCTCCAACATGAGCCACTTGGCGAG (mutated)

G → A mutation *DNMT3A* c.2645G>A

This one-letter change alters amino acid 882 in DNMT3A protein (hence ‘R882H’), frequently associated with acute myeloid leukaemia (AML).

Which blood cells contain this changed sequence in their RNA?

Single-cell DNA sequencing measures genotype. RNA sequencing also measures transcriptional state.

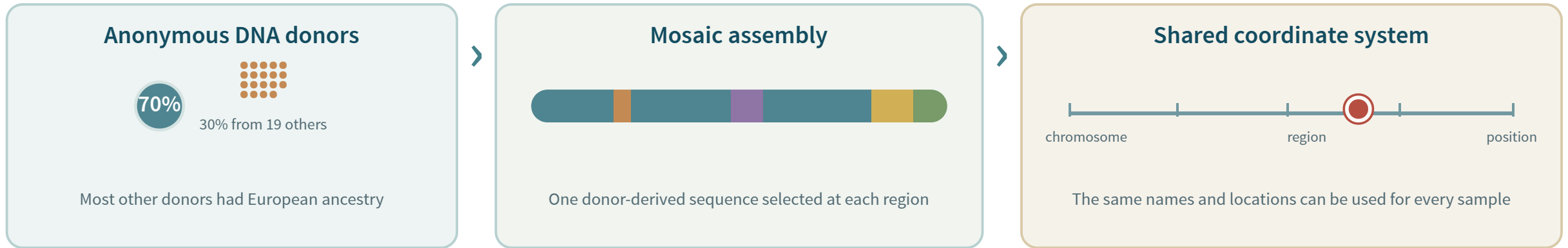
But we may also want to know:

- Which blood-cell types show the mutation?
- Do cells with the mutation form a distinct group, say transcriptionally?
- Do sequence-positive cells show different proliferation programs?

Glossary: reference genome

‘Alignment’: aligned to what?

Example: Human Genome Project; Sanger sequencing, reads of 500–1000 bps.



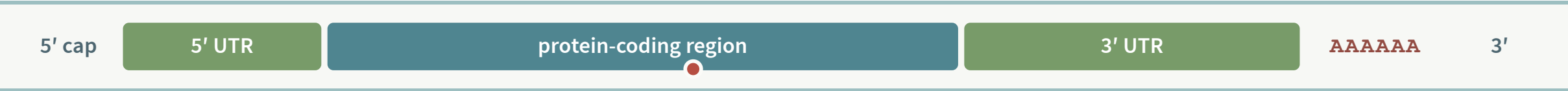
Reference sequence: the agreed sequence used to assign locations and compare observed DNA or RNA.

- The reference is a coordinate standard. It is not an average genome and does not preserve every donor's variants.
- Typical short sequencing reads: 50–300 bases.

Glossary: transcript ends and end-biased scRNA-seq

5' marks the beginning of an RNA transcript. 3' marks the end that carries the poly(A) tail.

UTR: untranslated region



5' protocol

reads near the transcript start



3' protocol

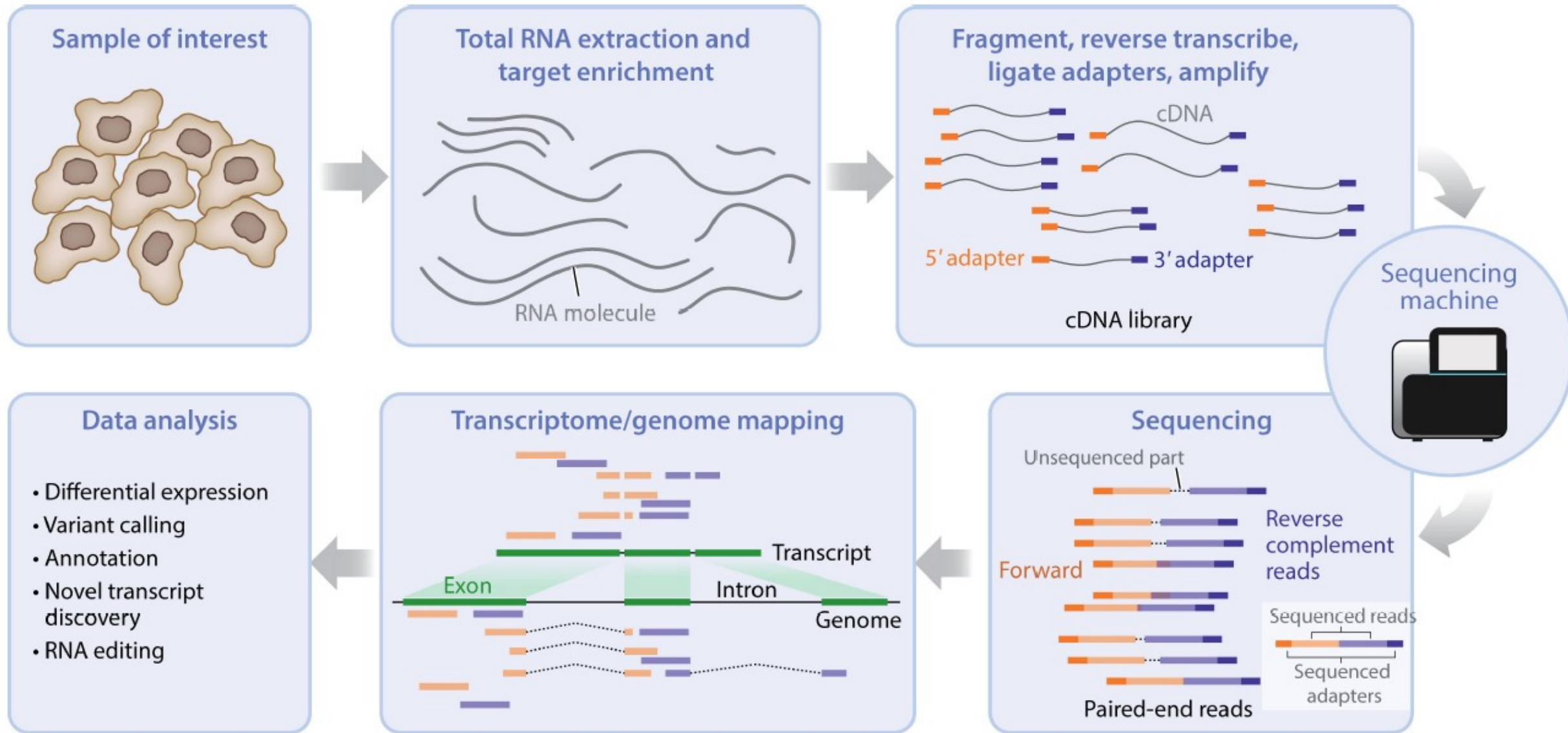
reads near the poly(A) end

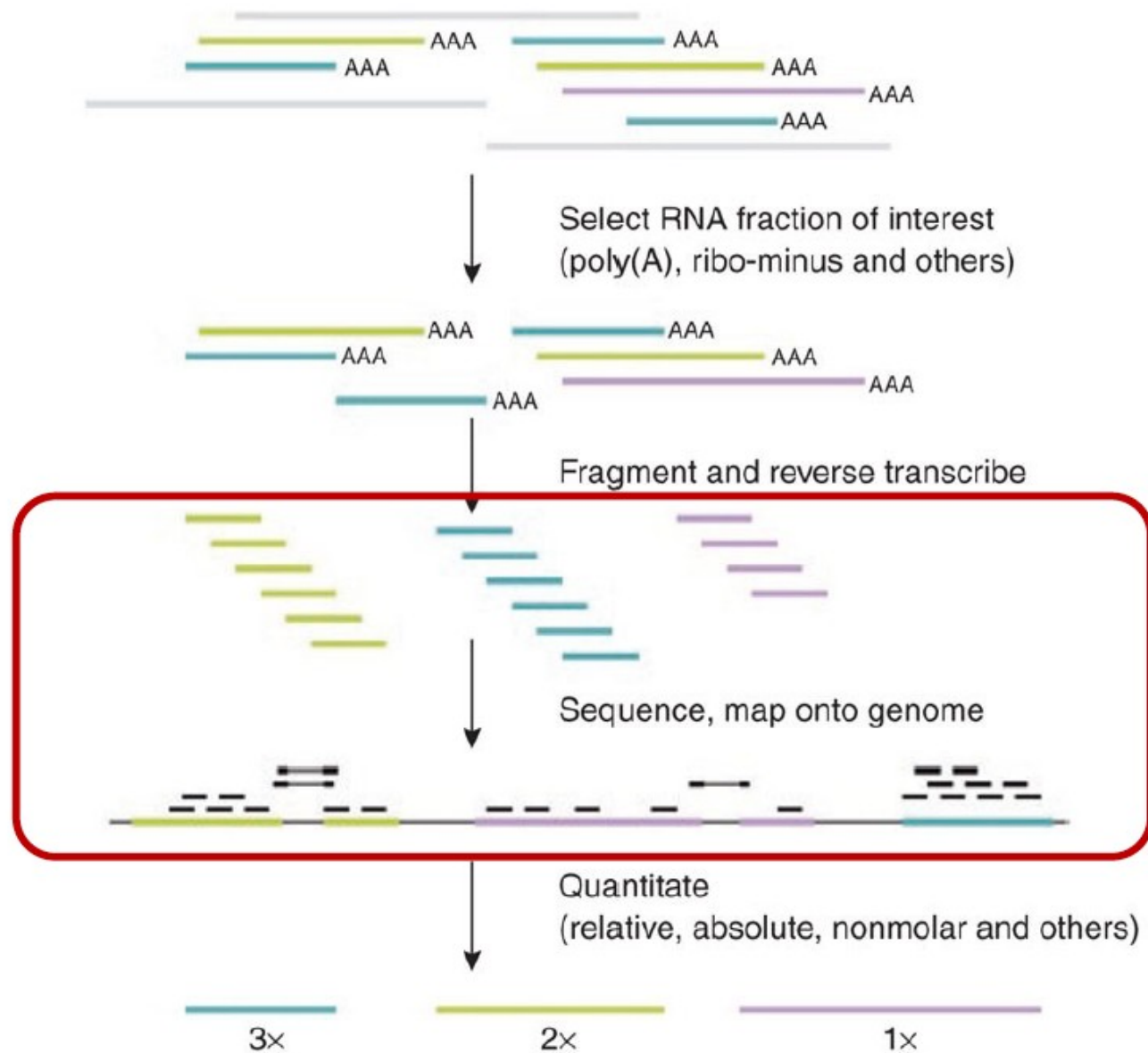


Coverage consequence

A gene may be detected while an internal mutation or exon receives no reads.

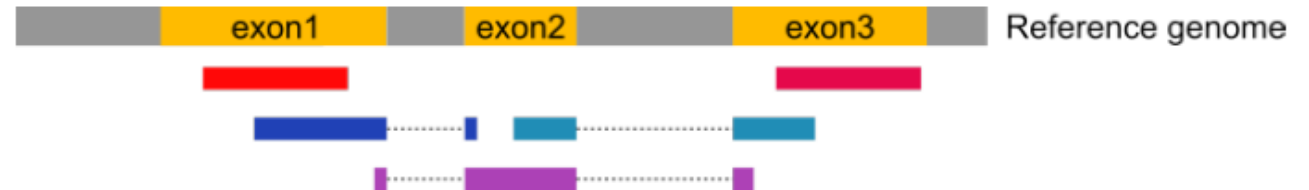
Xiaochen's part



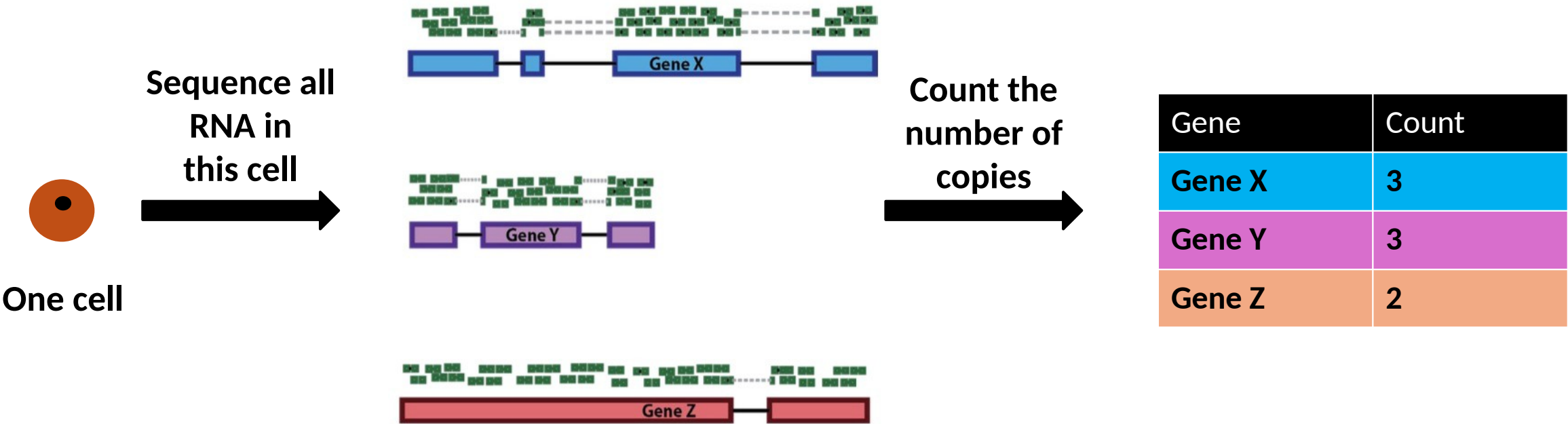


Alignment to a reference genome

- Assign every read to its location in the reference genome
- In eukaryotes, mRNA has introns spliced out
 - Reads might span splice junctions in reference genome
- Splice-aware aligner that is aware of introns
 - Popular tools include:
 - HISAT2 (Kim et al. 2015)
 - STAR (Dobin et al. 2013)
 - Subread (Liao et al. 2013)



Raw data processing: from short reads to counts



Different sequences can give the same count

Two cells, one gene, different sequence composition

Reference fragment: ACGTTGCA

Alternative fragment: ACATTGCA

Cell	Reference	Alternative	Gene X total
C1	3	0	3
C2	1	2	3

C1 and C2 both report Gene X = 3

The matrix cannot recover the original 3:0 and 1:2 split.

What is lost in a gene-count matrix?

Gene totals hide which sequences contributed to the count

Sequence detail	Question hidden by gene totals
Base identity	Which reference or alternative allele?
Exons and junctions	Which splice form or circular junction?
Transcript ends	Which 3' UTR or poly(A) site?
Sequences outside the feature set	Which virus, microbe, vector or novel RNA?

The matrix retains abundance per annotated gene and cell.

Retained reads or BAM files can still contain the missing evidence.

Malva changes the stored representation

The same sequencing data support different questions

Stage	Gene-count workflow	Malva sequence workflow
Input	RNA reads, barcode and UMI	The same sequencing data
Processing	Reference alignment Gene assignment and UMI collapsing	Sample k-mers Store their cell associations
Stored result	Cell × gene UMI counts	Sequence-to-cell index
Question	How much of gene X?	Which cells contain sequence Q?

For sequence queries, Malva bypasses reference-based gene quantification.

Gene counts remain useful. A Malva index is not a lossless FASTQ archive.

Building a sequence-to-cell index

A small worked example using $k = 4$

Distinct read types and their cells

Cell	Read split into 4-mers
C1	ACGT TGCA
C2	ACGT TGCA
C2	ACAT TGCA

Stored k-mer / cell associations

4-mer	Cells containing it
ACGT	C1, C2
ACAT	C2
TGCA	C1, C2

The barcode preserves cell identity without a reference genome.

Repeated associations do not become molecule counts in this index.

Querying the sequence retrieves cell C2

The alternative fragment distinguishes cells with identical gene totals

QUERY

ACAT

Alternative sequence

INDEX ENTRY

4-mer	Cells
ACAT	C2

MATCHED CELL

C2

Plus sample and
cell metadata

Exact k-mer lookups identify matching cells.

Longer probes use sliding windows and match thresholds.

Queries read relevant index blocks instead of rescanning FASTQ.

Sequence queries produce task-specific outputs

Choose the feature after indexing the reads

Query or feature	Output	Typical use
Known gene sequence	Cell × gene pseudocounts	Expression exploration
Allele, junction or UTR	Matched cells and sequence signal	Sequence-specific questions
K-mer composition	Cell × bucket features	Cell neighbours and clusters

Pseudocounts summarize sequence matches, not UMI molecule counts.

Gene names and biological interpretation may still use references.

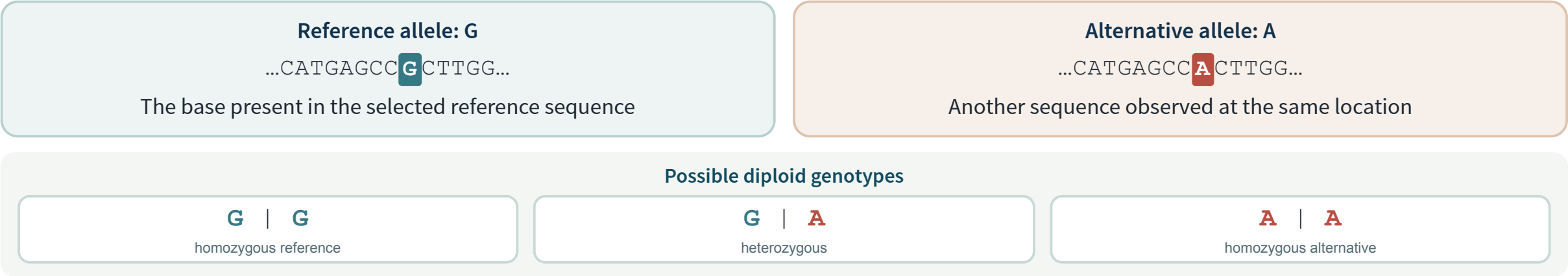
A missing hit can reflect limited RNA capture or coverage.

What's new?

- Sequence search at **single-cell resolution** across a large public index (searchable structure)
- Low-memory queries and incremental updates (new indexed datasets can be sequentially added)
- Cell-by-sequence analysis beyond targeted searches

Engineering tour de force: using conventional approaches we can probably do single-cell seq search on one dataset, not at this scale.

Glossary: allele, genotype and wild type



Allele	Genotype	Wild type
One sequence version at a genomic location	The alleles at a locus, usually two in a diploid cell	A context-dependent functional or conventional baseline

- The reference allele ≠ wild type
- If A is the functional baseline in a study, researchers may call A the wild-type allele.
- Example: sickle-cell; wild type: GAG → glutamate, mutant: GTG → valine

Glossary: variant, SNV and SNP

Variant

reference ...CGT...
observed ...CA**A**T...

Any sequence difference



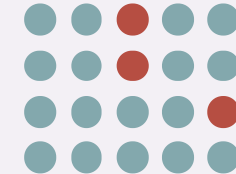
SNV

G → A

A single-nucleotide variant changes one base



SNP



An SNV present in $\geq 1\%$ of a population

Molecular description

SNV describes the size of the sequence difference.

Population description

SNP adds information about population frequency.

- Every SNP is an SNV. A rare SNV is not a SNP under the conventional 1% threshold.
- The 1% threshold is a convention and can vary between studies.

Glossary: germline, somatic and clonal

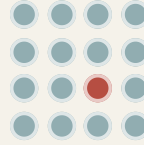
Germline variant



Present from conception and potentially throughout the body

Can be inherited

Somatic variant



Arises after conception in one cell

Restricted to that cell and its descendants

Clonal somatic variant



Variant descendants expand into a cell population

Example: clonal haematopoiesis

● cell without the variant ● cell carrying the variant

Germline screen Population variation

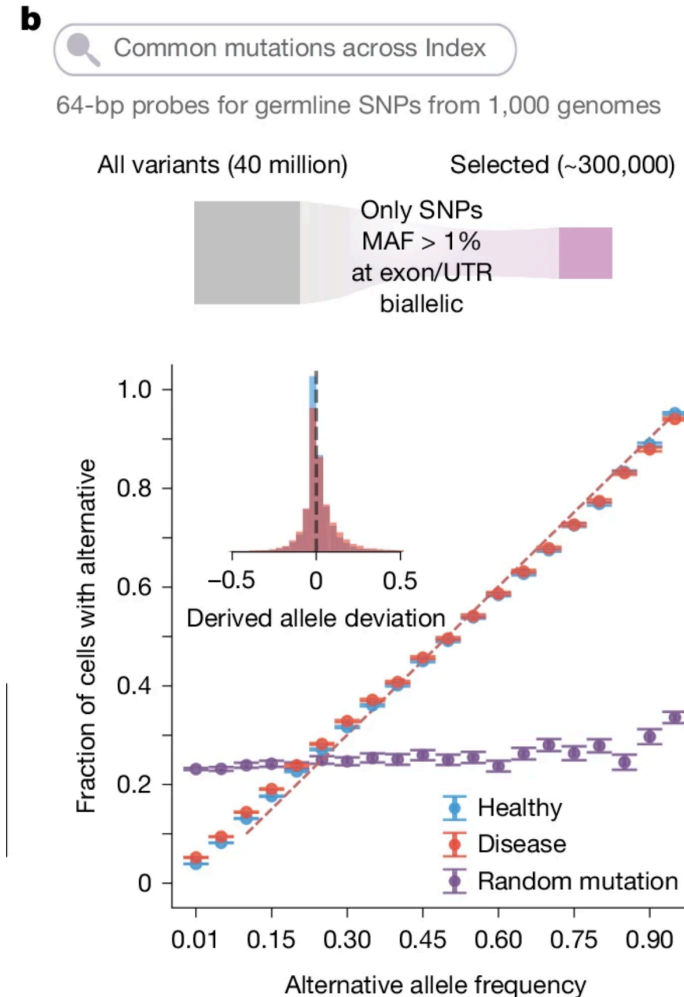
Somatic benchmark Cancer mutations

- SNV describes a one-base change; germline or somatic describes its origin.
- Germline: BRCA1 variant (breast cancer risk).
- Somatic: some DNMT3A mutations give blood stem cells a growth advantage.
- Clonal: clonal expansion of DNMT3A in blood (1 clone accounting for a large proportion of blood cells).

Case study: Germline screening

Can atlas RNA data recover known population variants?

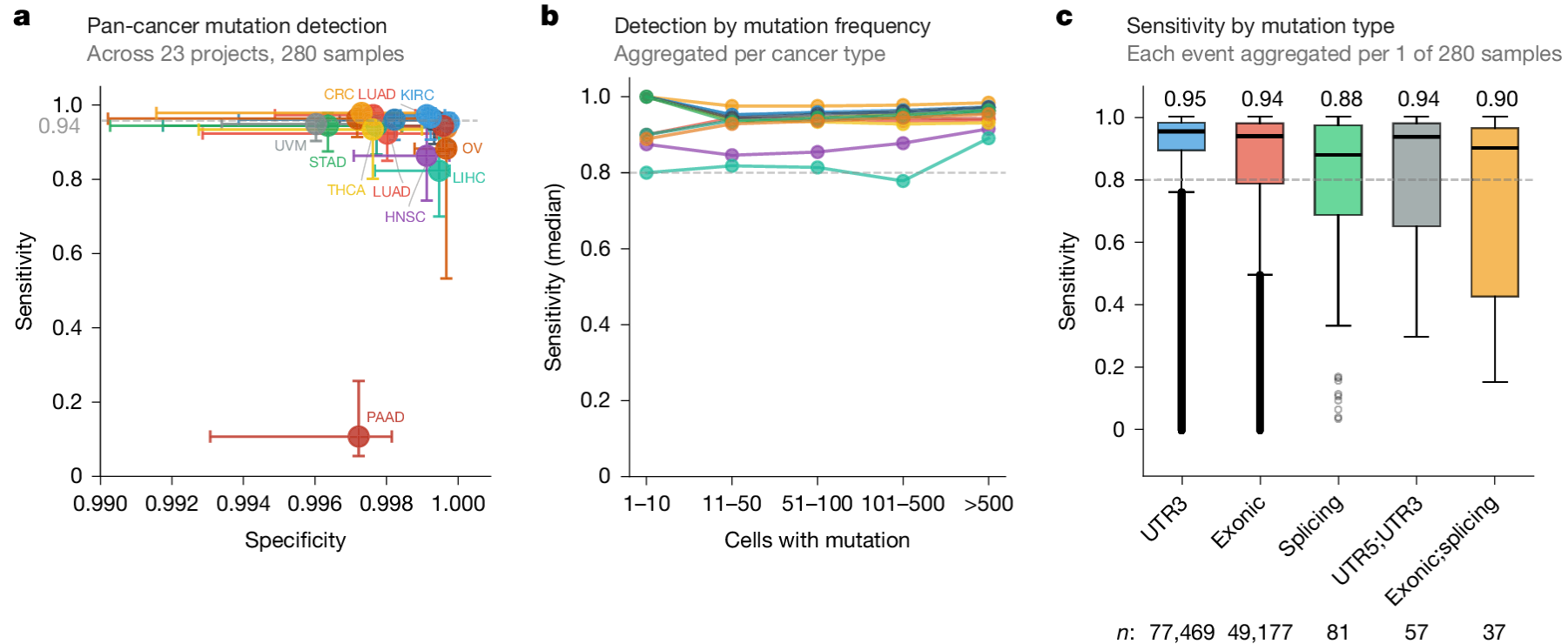
- 296,985 common, coding, biallelic SNPs
- 64-nucleotide allele-specific probes
- Healthy and disease samples
- Negative controls: probes with an extra altered base



Case study: Cancer somatic mutations

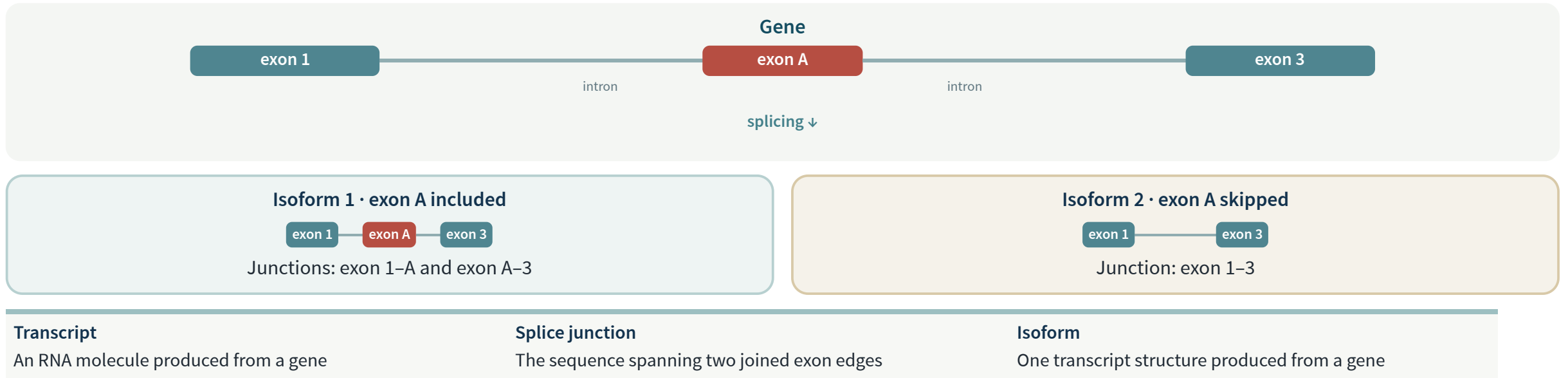
Can known tumour mutations be found in single-cell RNA?

- Allele-specific reference and alternative probes for expressed SNVs
- Comparison with scTML mutation calls
- 280 tumour samples across 16 cancer types



- PAAD outlier: mouse-derived samples were queried with human probes.
- Splice-region variants had lower median sensitivity than 3' UTR and exonic variants in this benchmark.
- The comparison uses supplied RNA-based mutation calls; a sequence hit does not establish DNA genotype.

Glossary: transcript, splice junction and isoform



- Two cells can have the same gene count but express different transcript isoforms.
- Two BCL2L1 splice forms illustrate this:
 - Bcl-xL suppresses cell death
 - Bcl-xS promotes cell death
- Alternative splicing may be due to splicing factors (proteins), DNA variants.

Case study: Isoform usage

Same gene

similar total counts

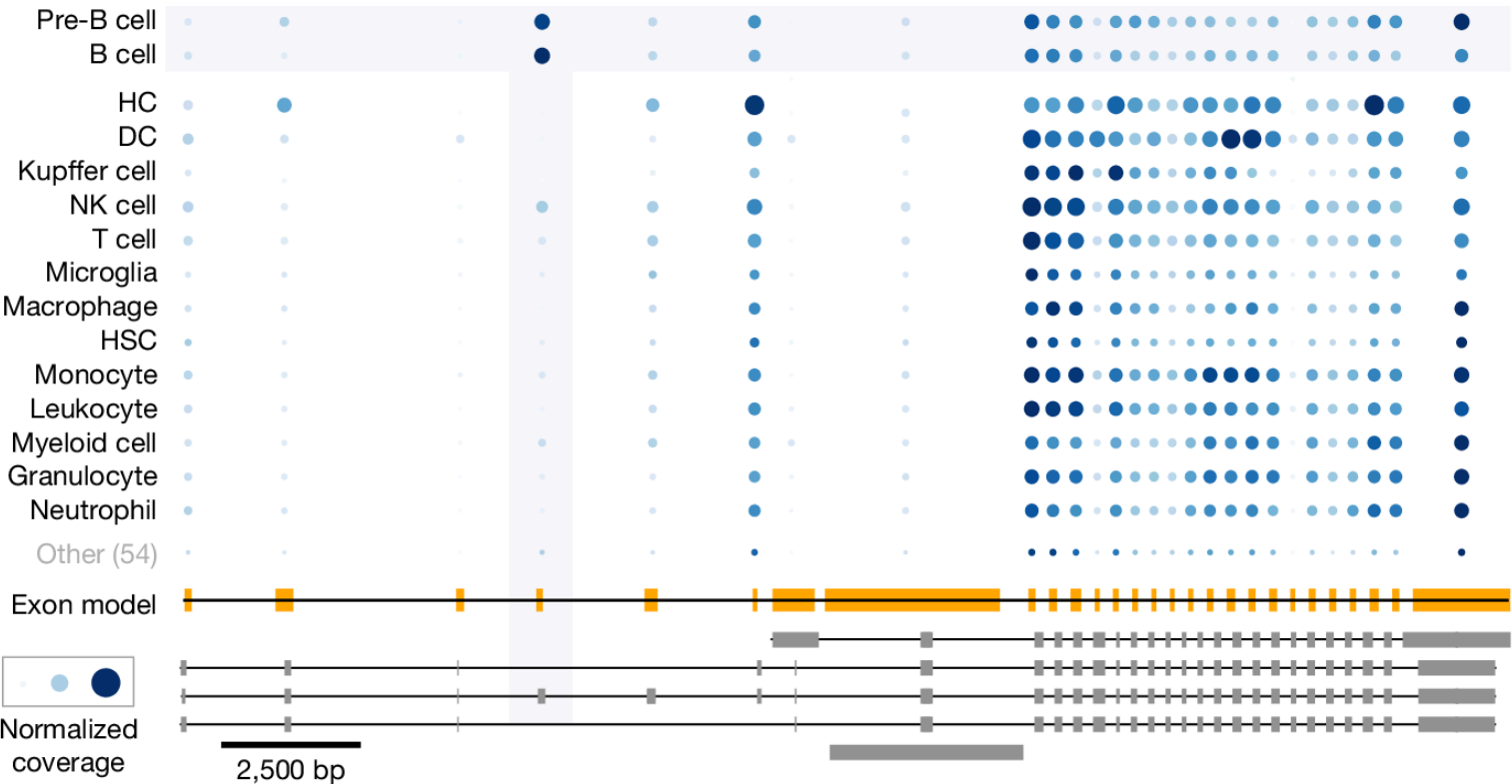
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Same transcript

different exons or 3' ends

- Probe a distinguishing exon, junction or 3' UTR segment
- Compare signal across cell types or spatial regions

Example: *Ptprc* exon A



B cells show stronger inclusion of exon A than the other immune populations.

Gene totals conceal this exon-level contrast.

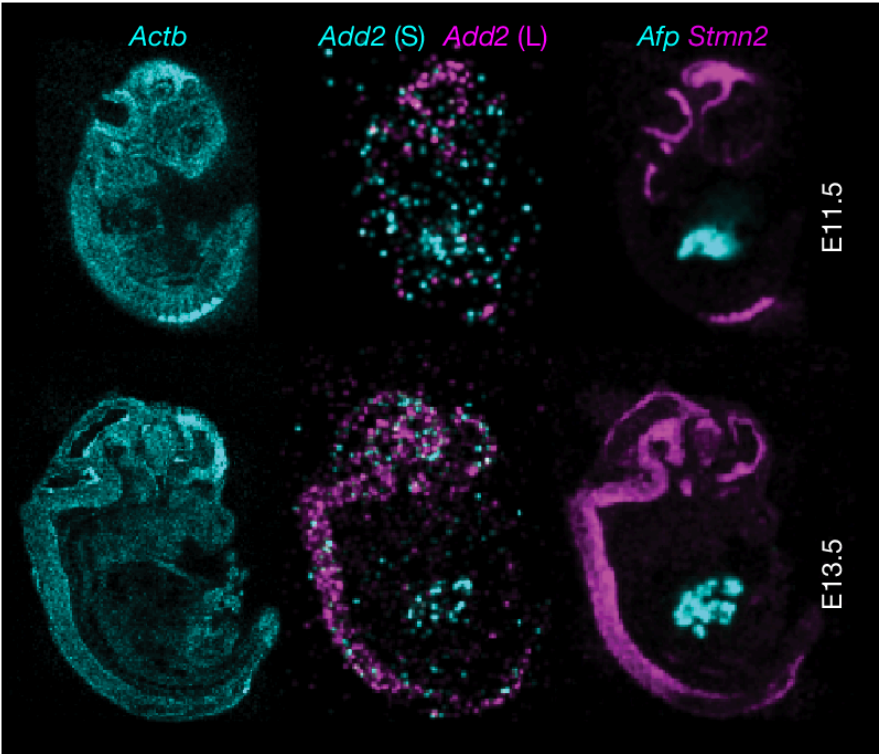
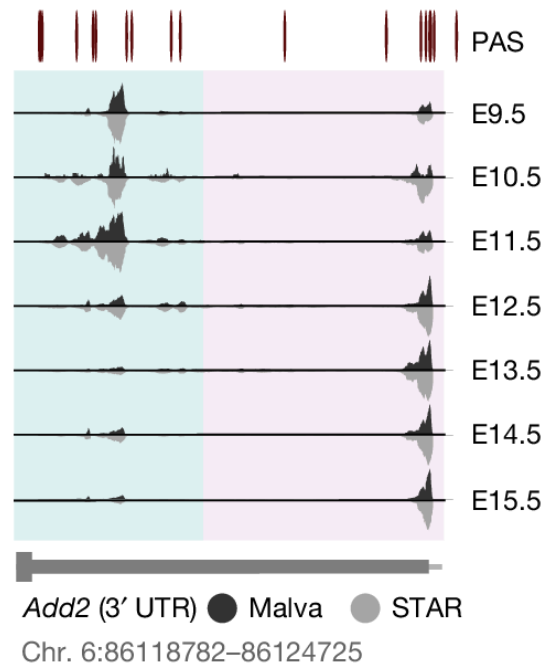
Example: *Add2* 3' UTR

d

Add2 3'UTR length in mouse embryo spatial atlas

Coverage-like query of 3' UTR and spatial projection

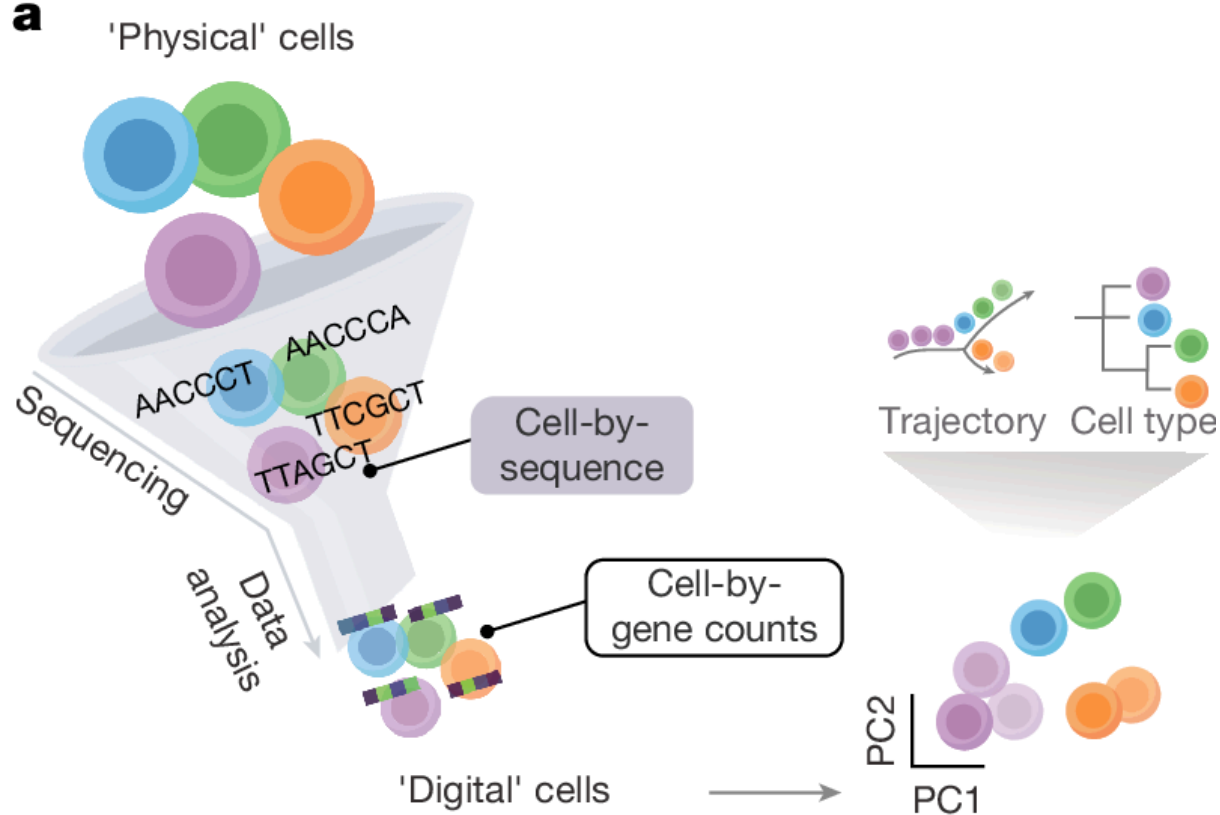
MOSTA atlas (Stereo-seq)



The distal 3' UTR signal localises to developing neural tissue; the proximal form accumulates in liver.

Spatial context explains the apparent developmental shift in bulk profiles.

Case study: From sequence search to an assay

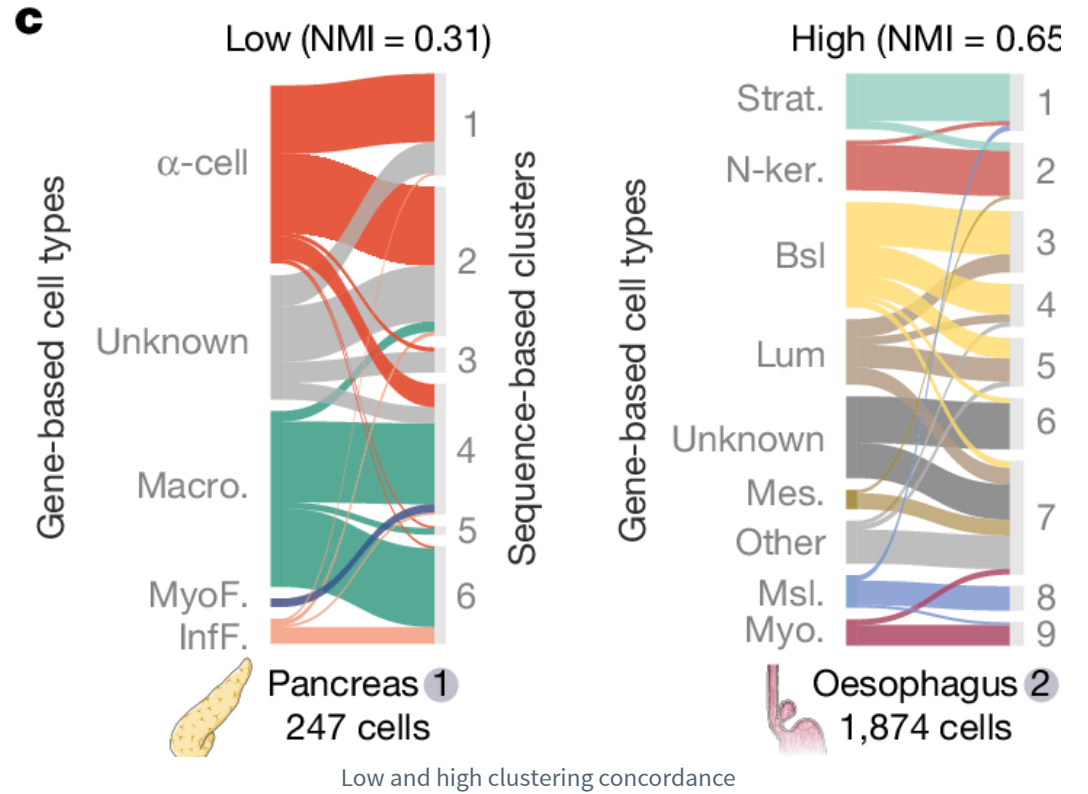
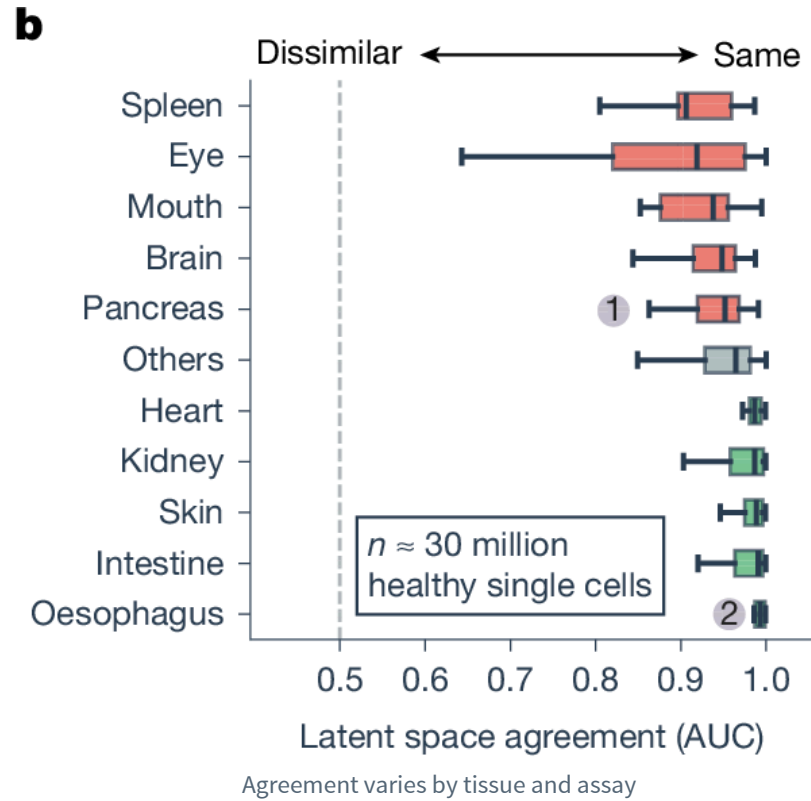


Most examples so far validate known patterns.

The same reads can support a gene-count representation or a sequence-derived representation.

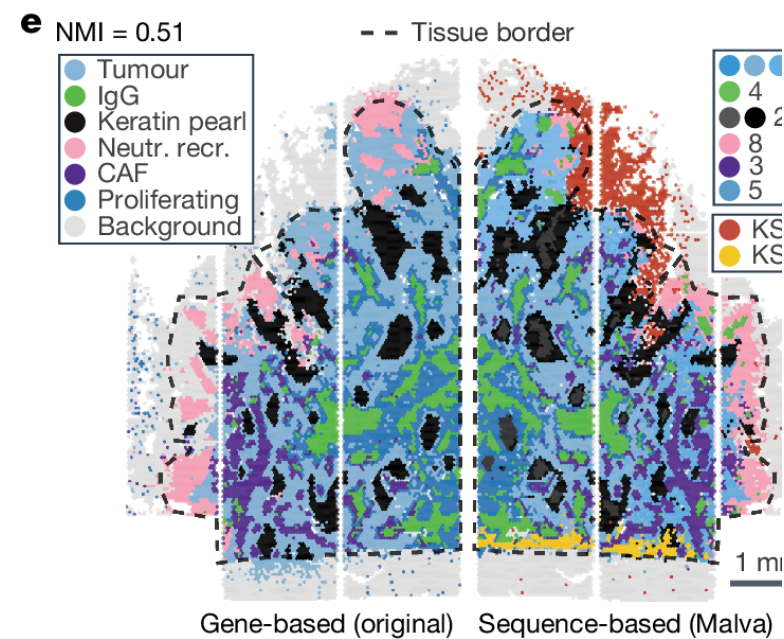
- Malva groups related k-mers into buckets, then builds an embedding for neighbours and clusters.
- This opens up possibilities for **discovery**.

Cell clustering from sequence composition

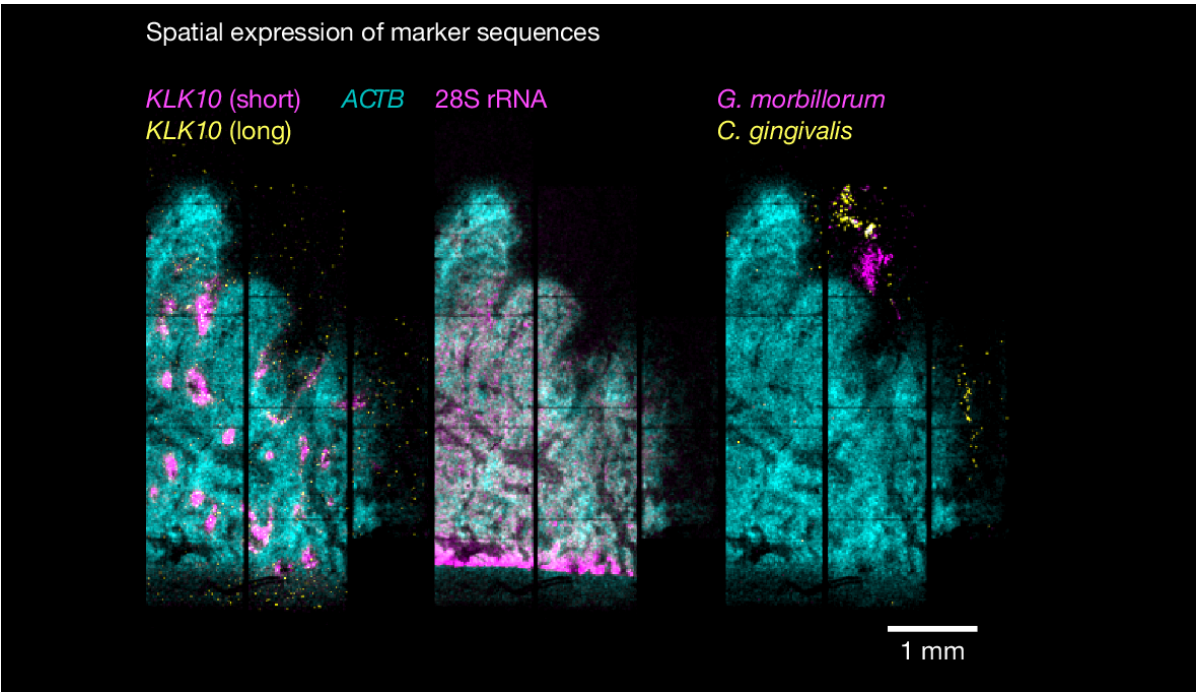


- Clustering buckets vs clustering counts
- Pancreas: disagreement; Oesophagus: agreement

From a cluster to a marker sequence



Sequence-derived clusters preserve tissue structure



Assembled markers localise within and outside the tissue

KLK10 short-form signal localised within tumour regions; bacterial rRNA markers appeared outside the tissue.

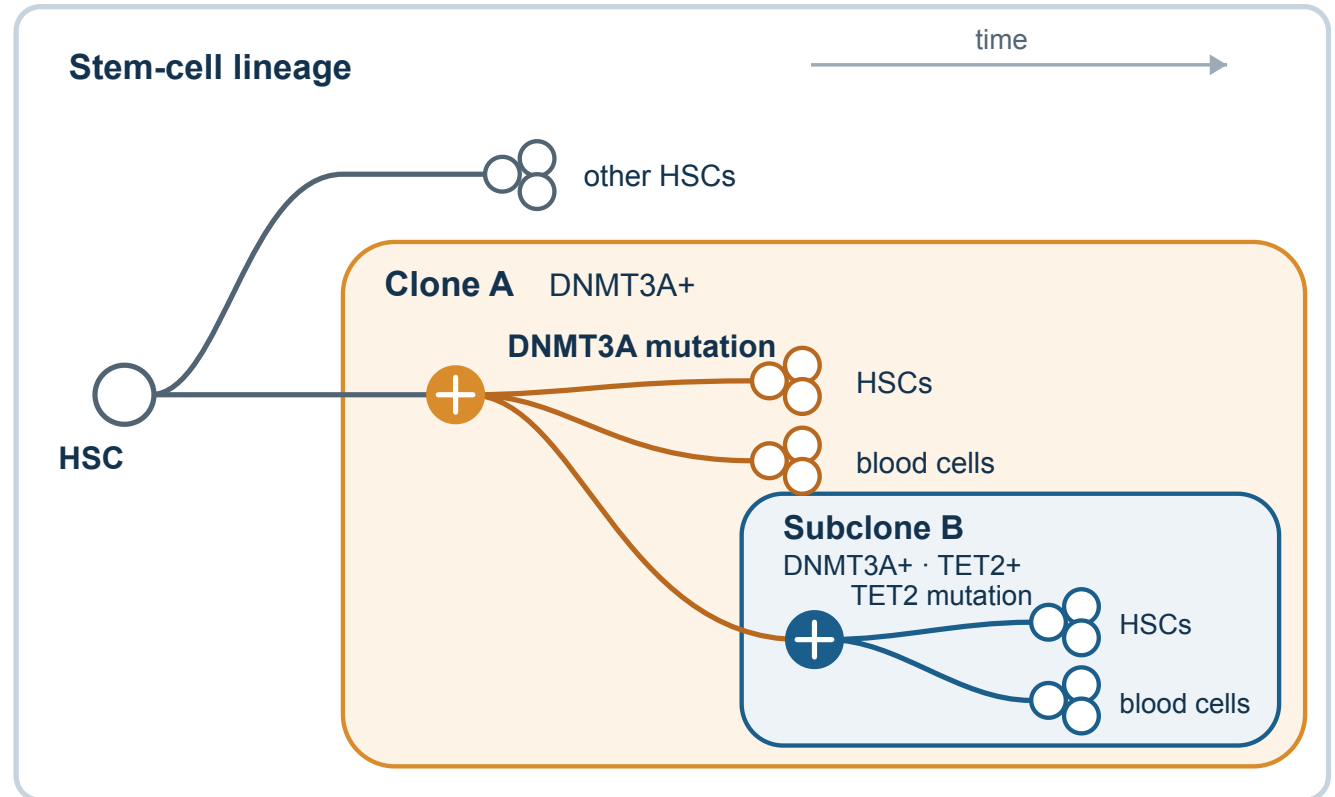
What can we do with the tool?

What MALVA returns

- Sequence → matching cells and samples
- Coverage along a transcript or region
- Sequence buckets → clusters and markers

Possible uses

- Survey clonal haematopoiesis in Human Cell Atlas data
- Sequence features alongside gene counts – buy one get one free?
- How reliably can RNA hits identify clones and subclones?



Limitations

scRNA-seq cannot cover

- DNA that is not transcribed, such as most intergenic regions, promoters and enhancers;
- genes not expressed in the sampled cell;
- most introns in mature messenger RNA;
- lowly expressed transcripts, because sampling is sparse;
- positions far from the captured transcript end in 3'- or 5'-biased protocols;
- an allele that is not expressed.

Example: DNMT3A R882H hard to capture using 3' methods

But do you want the free bubble tea?

References

- León-Periñán D, Karaiskos N, Rajewsky N. *Ultrafast and reference-free sequence discovery in single-cell data*. Nature. 2026.
<https://doi.org/10.1038/s41586-026-10975-w>
- [Malva client and examples](#)
- [Malva local tools documentation](#)

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